

(19) **United States**
 (12) **Patent Application Publication** (10) **Pub. No.: US 2025/0280005 A1**
Soryal (43) **Pub. Date: Sep. 4, 2025**

(54) **METHODS, SYSTEMS, AND DEVICES FOR VERIFYING A LOCATION OF A COMMUNICATION DEVICE**

(52) **U.S. Cl.**
 CPC *H04L 63/107* (2013.01); *H04L 63/0272* (2013.01); *H04L 63/0492* (2013.01); *H04L 63/08* (2013.01)

(71) Applicant: **AT&T Intellectual Property I, L.P.**,
 Atlanta, GA (US)

(72) Inventor: **Joseph Soryal**, Glendale, NY (US)

(73) Assignee: **AT&T Intellectual Property I, L.P.**,
 Atlanta, GA (US)

(21) Appl. No.: **19/208,713**

(22) Filed: **May 15, 2025**

Related U.S. Application Data

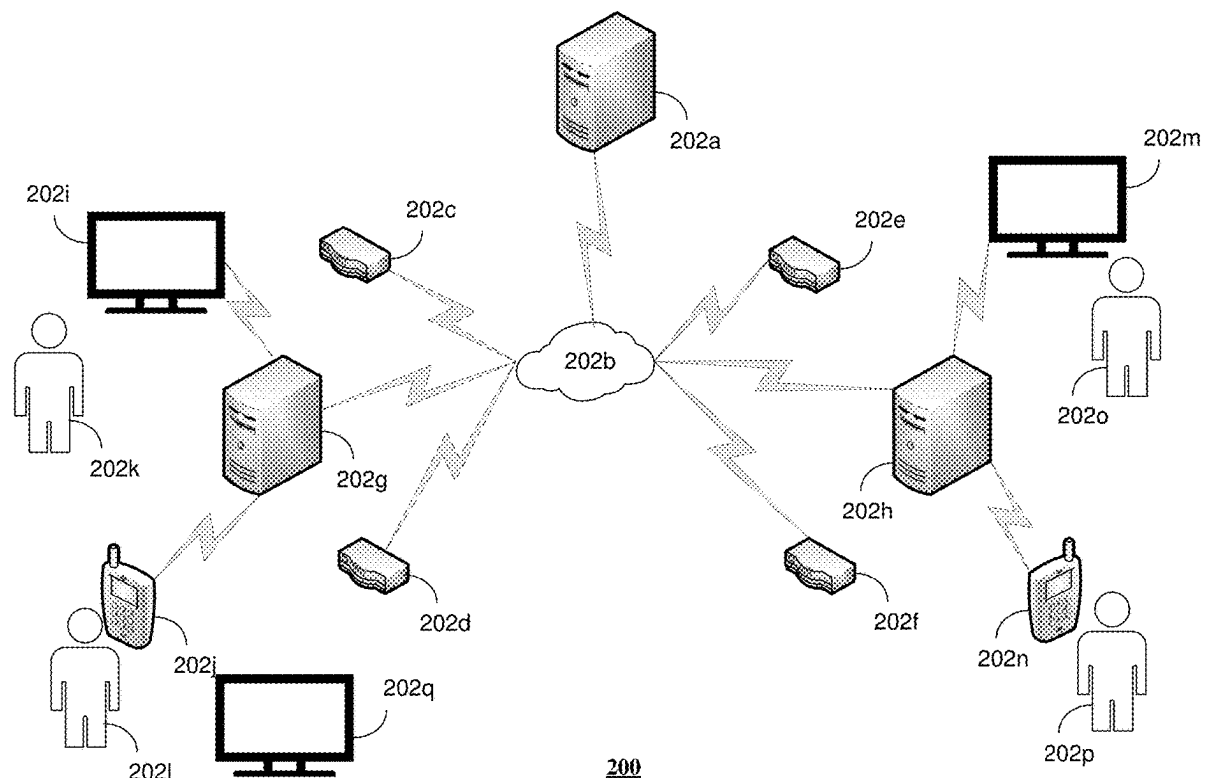
(63) Continuation of application No. 18/764,792, filed on Jul. 5, 2024, now Pat. No. 12,323,430, which is a continuation of application No. 17/573,831, filed on Jan. 12, 2022, now Pat. No. 12,052,264.

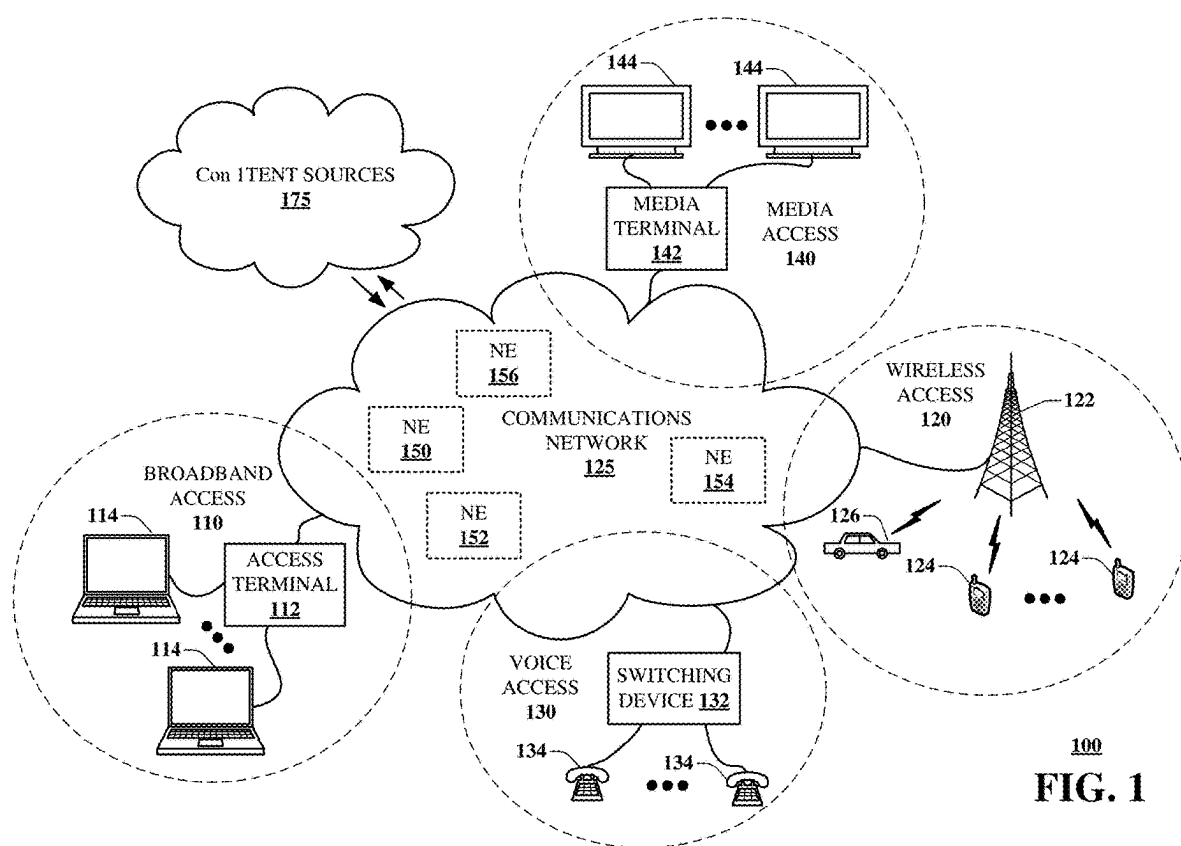
Publication Classification

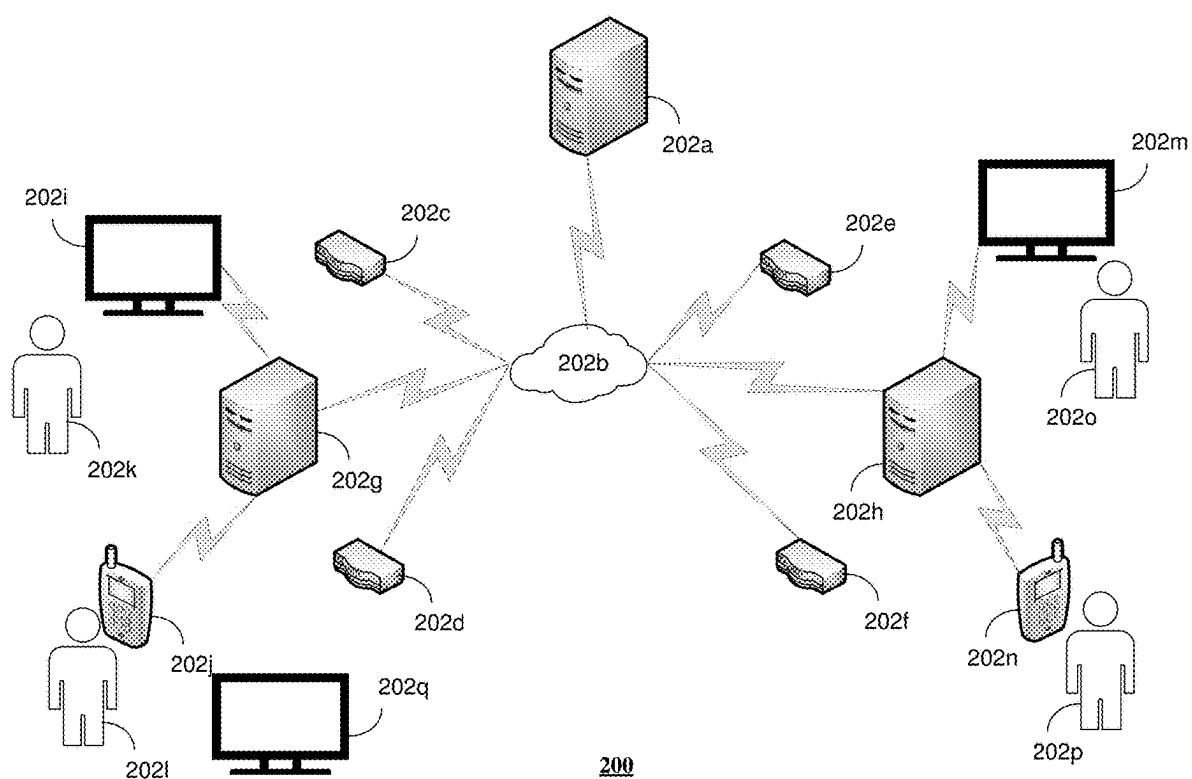
(51) **Int. Cl.**
H04L 9/40 (2022.01)

(57) ABSTRACT

Aspects of the subject disclosure may include, for example, a processing system determining a first communication device initiating a communication session with a second communication device, and transmitting a notification to the second communication device. The notification indicates to the second communication device to initiate a location verification application on the second communication device, and the location verification application transmits notifications to adjacent network devices. Each notification indicates to an adjacent network device to provide its location to the processing system. Further embodiments include receiving locations from the adjacent network devices, and identifying a first location associated with the second communication device. Additional embodiments include determining that the first location is in proximity to the locations, and establishing the communication session between the first communication device and the second communication device. Other embodiments are disclosed.







200
FIG. 2A

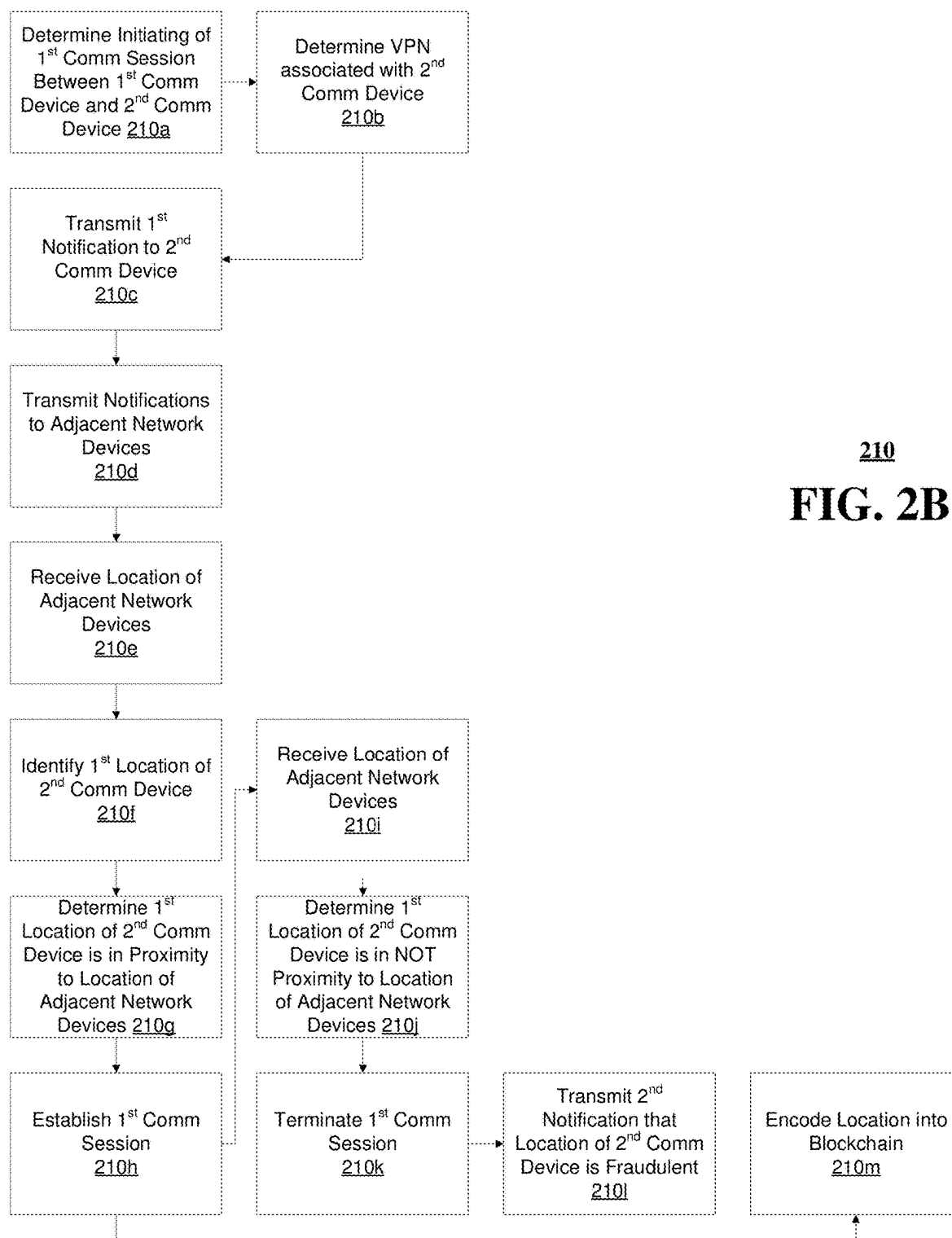
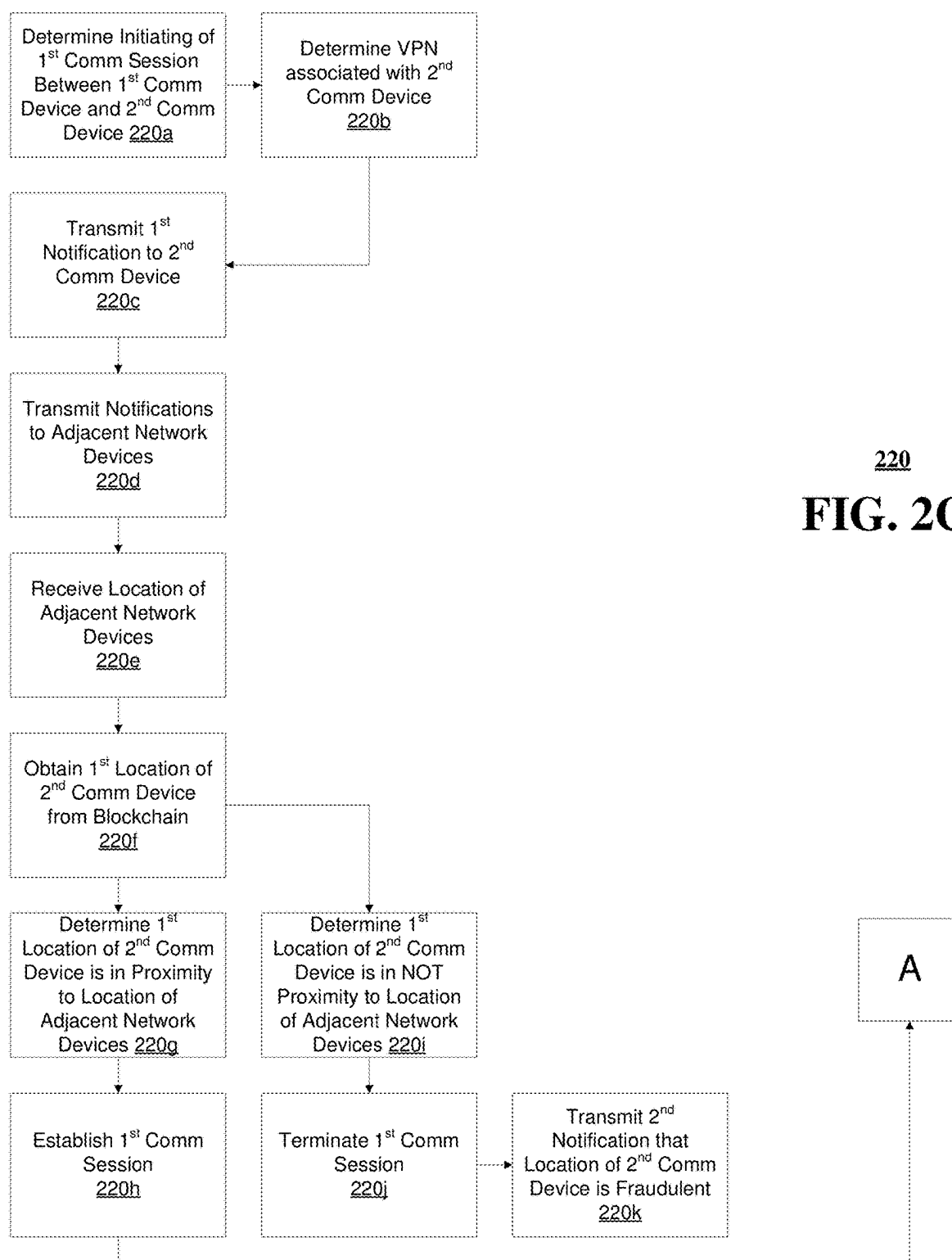
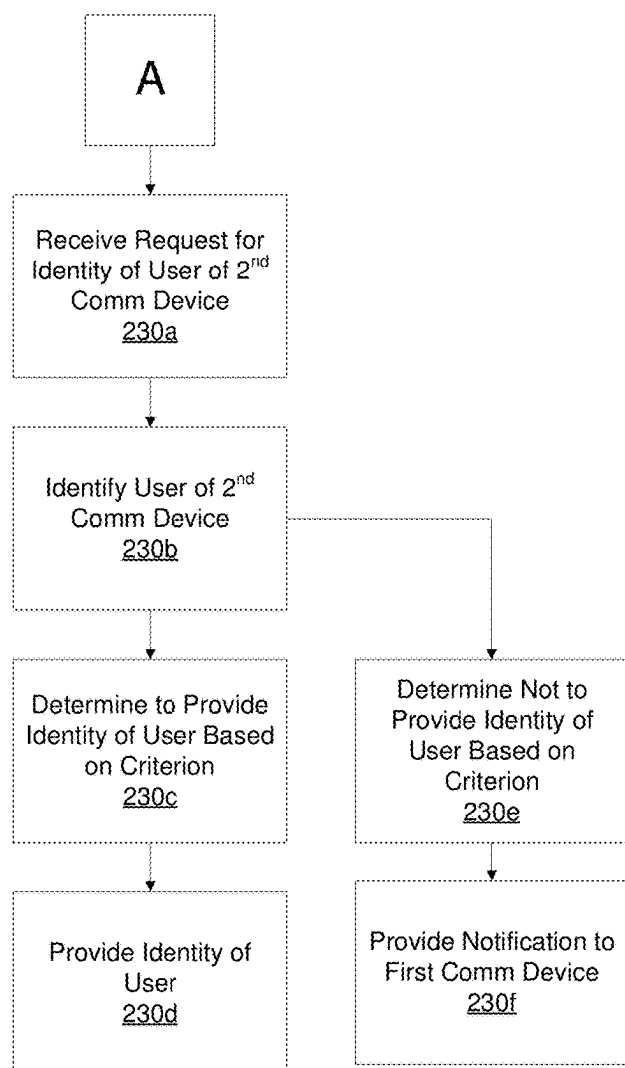
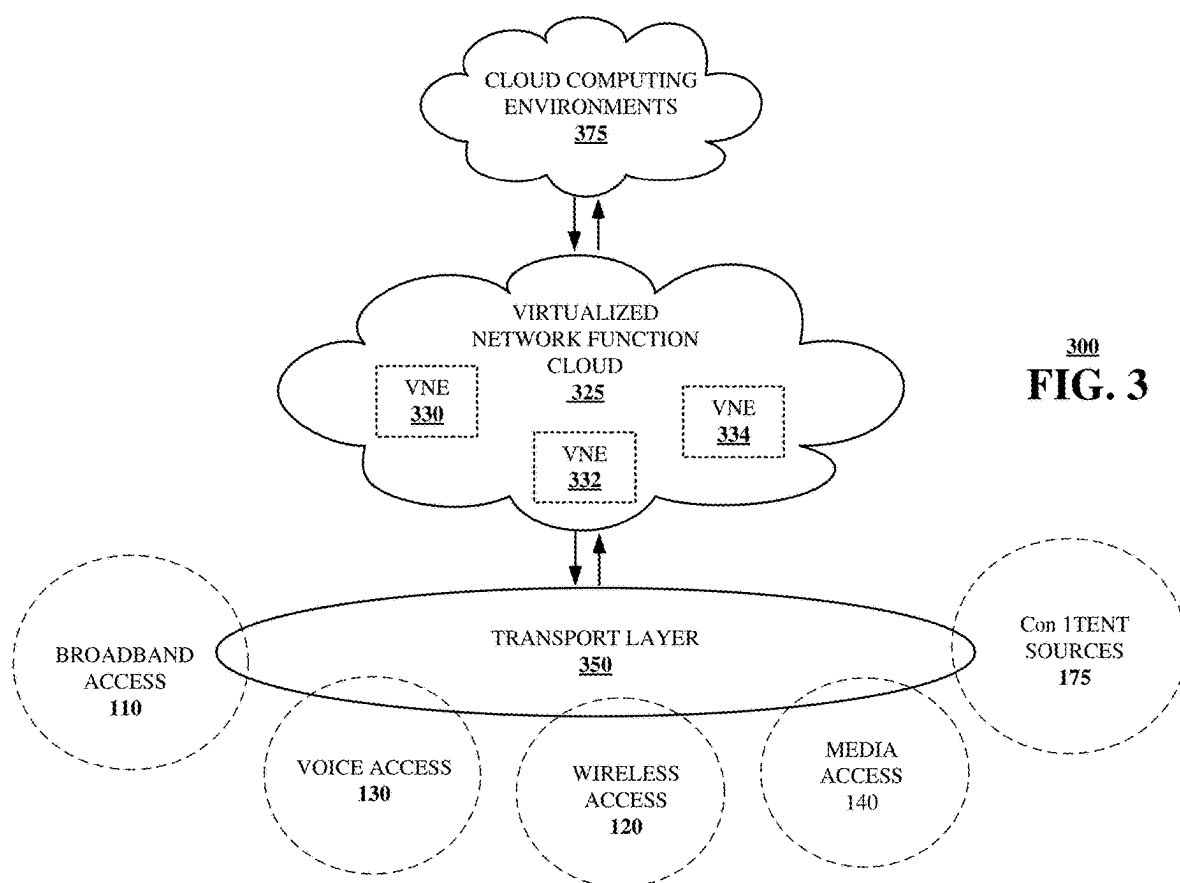


FIG. 2B





230
FIG. 2D



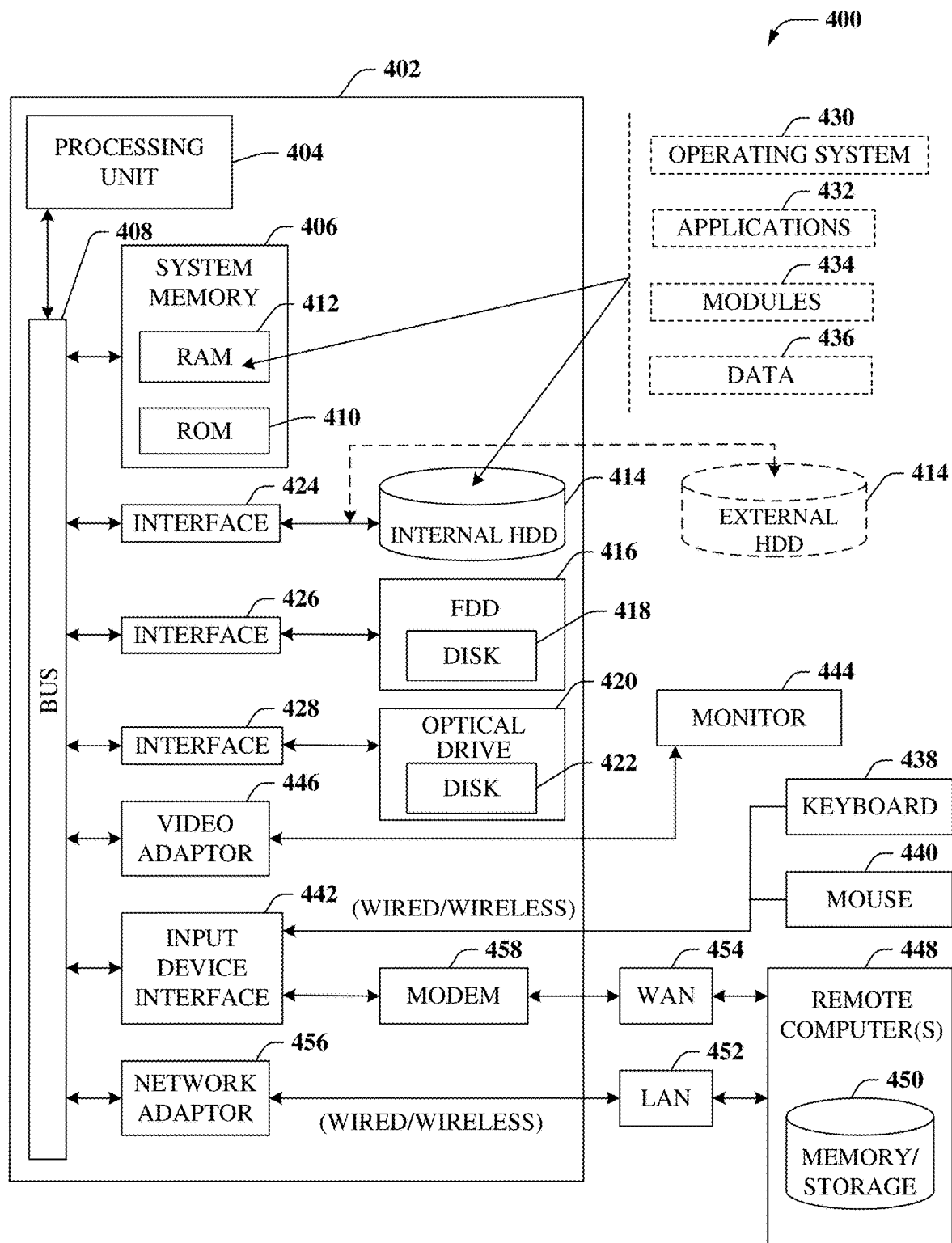


FIG. 4

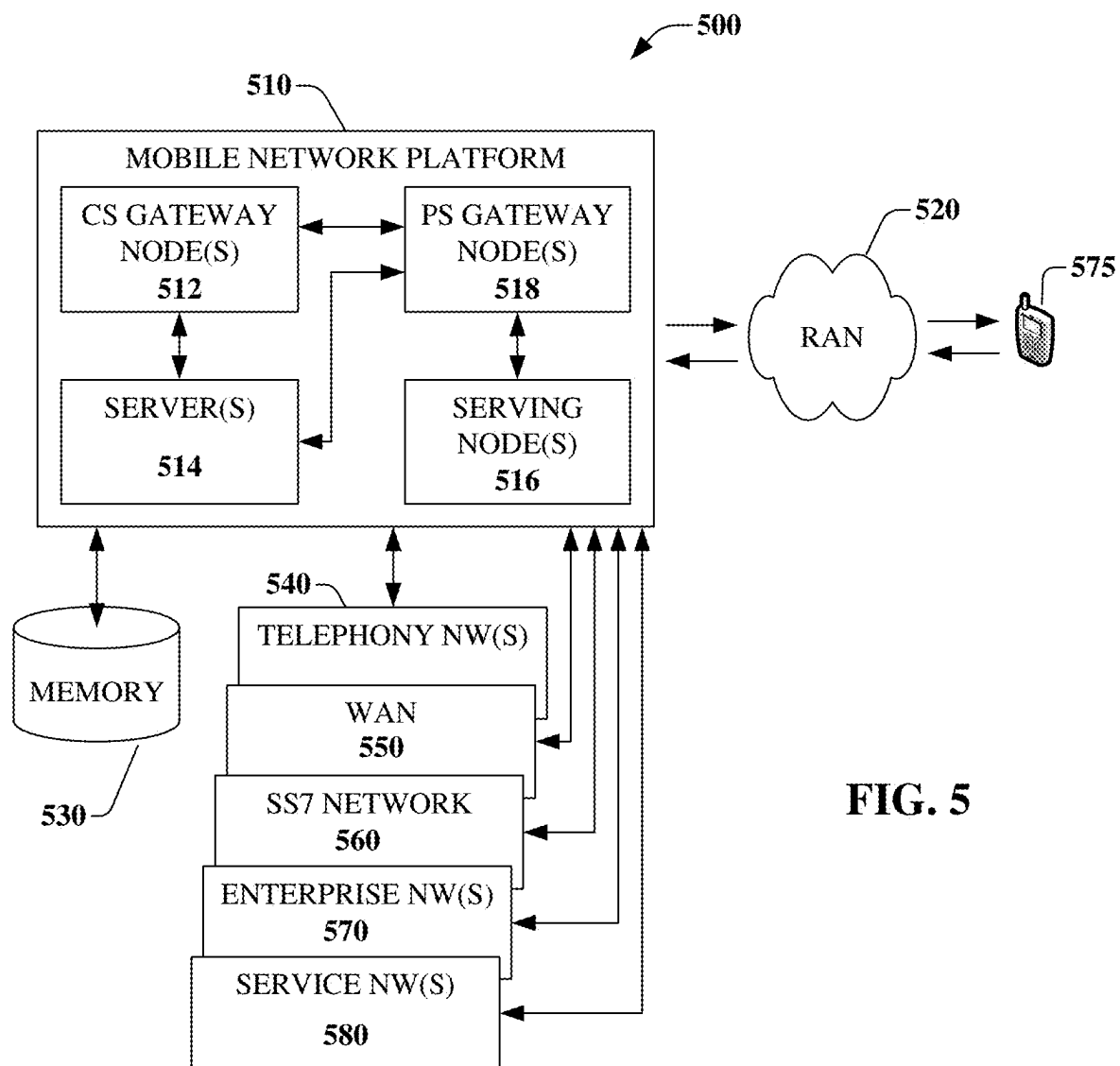
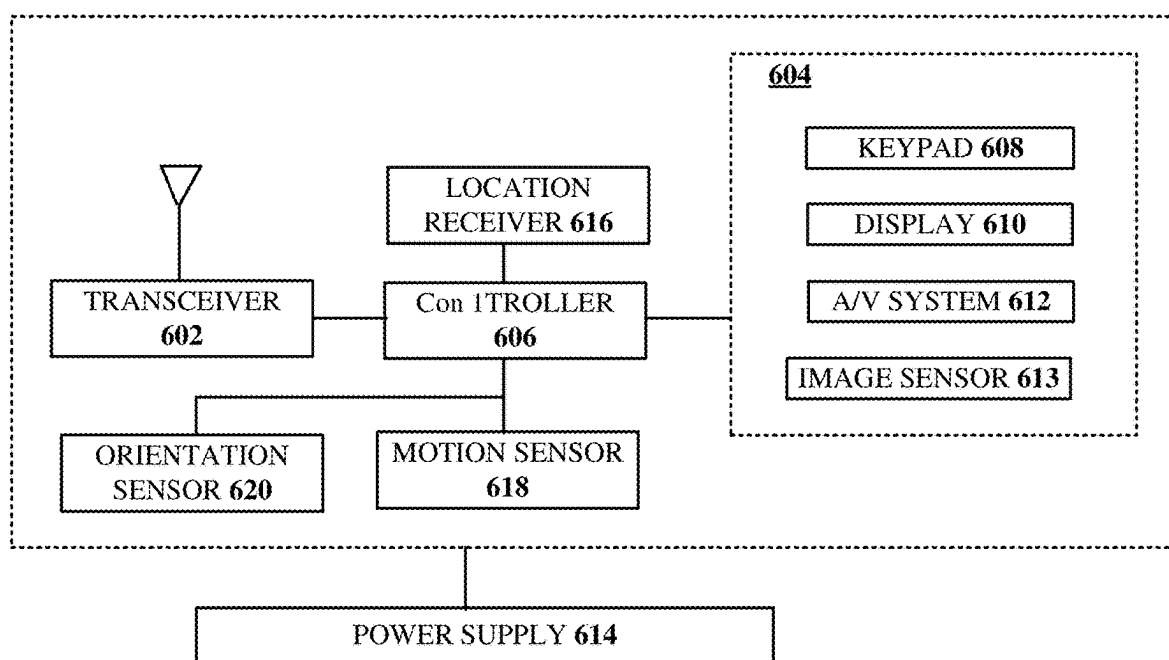


FIG. 5



600
FIG. 6

METHODS, SYSTEMS, AND DEVICES FOR VERIFYING A LOCATION OF A COMMUNICATION DEVICE

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is a continuation of U.S. patent application Ser. No. 18/764,792, filed Jul. 5, 2024, which is a continuation of U.S. patent application Ser. No. 17/573,831, filed Jan. 12, 2022, now U.S. Pat. No. 12,052,264. All sections of the aforementioned application(s) and/or patent (s) are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The subject disclosure relates to methods, systems, and devices for verifying a location of a communication device.

BACKGROUND

[0003] Current state of the art includes initiating and establishing a communication session between two communication devices, each associated with a different user. The communication session can be an audio-only communication session or a video communication session between the users. In certain instances, the two users may not know each other and a first user may require verification of location of the second user to authenticate the user (e.g., for a discussion of finances, benefits, medical issues, etc.). However, the communication device of the second user may utilize a virtual private network (VPN) that can mask the location of the communication device of the second user prohibiting authentication.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0005] FIG. 1 is a block diagram illustrating an exemplary, non-limiting embodiment of a communications network in accordance with various aspects described herein.

[0006] FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a system functioning within the communication network of FIG. 1 in accordance with various aspects described herein.

[0007] FIGS. 2B-2D depicts illustrative embodiments of methods in accordance with various aspects described herein.

[0008] FIG. 3 is a block diagram illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein.

[0009] FIG. 4 is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.

[0010] FIG. 5 is a block diagram of an example, non-limiting embodiment of a mobile network platform in accordance with various aspects described herein.

[0011] FIG. 6 is a block diagram of an example, non-limiting embodiment of a communication device in accordance with various aspects described herein.

DETAILED DESCRIPTION

[0012] The subject disclosure describes, among other things, illustrative embodiments for determining a first communication device initiating a first communication session with a second communication device over a communication network, and transmitting a first notification to the second communication device. The first notification indicates to the second communication device to initiate a location verification application on the second communication device. The location verification application transmits a group of notifications to a group of adjacent network devices. Each notification of the group of notifications indicates to an adjacent network device of the group of adjacent network devices to provide its location to the processing system. Further embodiments can include receiving a first group of locations from the group of adjacent network devices over the communication network, and identifying a first location associated with the second communication device. Additional embodiments can include determining that the first location is in proximity to the first group of locations, and establishing the first communication session between the first communication device and the second communication device over the communication network. Other embodiments are described in the subject disclosure.

[0013] One or more aspects of the subject disclosure include a device, comprising a processing system including a processor, and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations. The operations can comprise determining a first communication device initiating a first communication session with a second communication device over a communication network, and transmitting a first notification to the second communication device. The first notification indicates to the second communication device to initiate a location verification application on the second communication device. The location verification application transmits a group of notifications to a group of adjacent network devices. Each notification of the group of notifications indicates to an adjacent network device of the group of adjacent network devices to provide its location to the processing system. Further operations can include receiving a first group of locations from the group of adjacent network devices over the communication network, and identifying a first location associated with the second communication device. Additional operations can include determining that the first location is in proximity to the first group of locations, and establishing the first communication session between the first communication device and the second communication device over the communication network.

[0014] One or more aspects of the subject disclosure include a non-transitory, machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations. The operations can comprise determining a first communication device initiating a first communication session with a second communication device over a communication network, and transmitting a first notification to the second communication device. The first notification indicates to the second communication device to initiate a location verification application on the second communication device. The location verification application transmits a group of notifications to a group of adjacent network devices. Each notification of the group of notifica-

tions indicates to an adjacent network device of the group of adjacent network devices to provide its location to the processing system. Further operations can include receiving a first group of locations from the group of adjacent network devices over the communication network, identifying a first location associated with the second communication device, determining that the first location is in proximity to the first group of locations, and establishing the first communication session between the first communication device and the second communication device over the communication network. Additional operations can include receiving a request for an identity of a user associated with the second communication device from the first communication device, identifying the user associated with the second communication device resulting in an identification of the user, determining not to provide the identity of the user to the first communication device based on a criterion and the identification resulting in a determination, and providing a second notification to the first communication device that indicates the identity of the user cannot be revealed based on the determination.

[0015] One or more aspects of the subject disclosure include a method, comprising determining, by a processing system including a processor, a first communication device initiating a first communication session with a second communication device over a communication network, and transmitting, by the processing system, a first notification to the second communication device. The first notification indicates to the second communication device to initiate a location verification application on the second communication device. The location verification application transmits a group of notifications to a group of adjacent network devices. Each notification of the group of notifications indicates to an adjacent network device of the group of adjacent network devices to provide its location to the processing system. Further, the method can comprise receiving, by the processing system, a first group of locations from the group of adjacent network devices over the communication network, and obtaining, by the processing system, a first location associated with the second communication device from a blockchain. In addition, the method can include determining, by the processing system, that the first location is in proximity to the first group of locations, and establishing, by the processing system, the first communication session between the first communication device and the second communication device over the communication network.

[0016] Referring now to FIG. 1, a block diagram is shown illustrating an example, non-limiting embodiment of a system 100 in accordance with various aspects described herein. For example, system 100 can facilitate in whole or in part verifying location of a communication device with the location of adjacent network devices to initiate a communication session. In particular, a communications network 125 is presented for providing broadband access 110 to a plurality of data terminals 114 via access terminal 112, wireless access 120 to a plurality of mobile devices 124 and vehicle 126 via base station or access point 122, voice access 130 to a plurality of telephony devices 134, via switching device 132 and/or media access 140 to a plurality of audio/display devices 144 via media terminal 142. In addition, communication network 125 is coupled to one or more content sources 175 of audio, video, graphics, text and/or other media. While broadband access 110, wireless access

120, voice access 130 and media access 140 are shown separately, one or more of these forms of access can be combined to provide multiple access services to a single client device (e.g., mobile devices 124 can receive media content via media terminal 142, data terminal 114 can be provided voice access via switching device 132, and so on).

[0017] The communications network 125 includes a plurality of network elements (NE) 150, 152, 154, 156, etc. for facilitating the broadband access 110, wireless access 120, voice access 130, media access 140 and/or the distribution of content from content sources 175. The communications network 125 can include a circuit switched or packet switched network, a voice over Internet protocol (VoIP) network, Internet protocol (IP) network, a cable network, a passive or active optical network, a 4G, 5G, or higher generation wireless access network, WIMAX network, UltraWideband network, personal area network or other wireless access network, a broadcast satellite network and/or other communications network.

[0018] In various embodiments, the access terminal 112 can include a digital subscriber line access multiplexer (DSLAM), cable modem termination system (CMTS), optical line terminal (OLT) and/or other access terminal. The data terminals 114 can include personal computers, laptop computers, netbook computers, tablets or other computing devices along with digital subscriber line (DSL) modems, data over coax service interface specification (DOCSIS) modems or other cable modems, a wireless modem such as a 4G, 5G, or higher generation modem, an optical modem and/or other access devices.

[0019] In various embodiments, the base station or access point 122 can include a 4G, 5G, or higher generation base station, an access point that operates via an 802.11 standard such as 802.11n, 802.11ac or other wireless access terminal. The mobile devices 124 can include mobile phones, e-readers, tablets, phablets, wireless modems, and/or other mobile computing devices.

[0020] In various embodiments, the switching device 132 can include a private branch exchange or central office switch, a media services gateway, VoIP gateway or other gateway device and/or other switching device. The telephony devices 134 can include traditional telephones (with or without a terminal adapter), VoIP telephones and/or other telephony devices.

[0021] In various embodiments, the media terminal 142 can include a cable head-end or other TV head-end, a satellite receiver, gateway or other media terminal 142. The display devices 144 can include televisions with or without a set top box, personal computers and/or other display devices.

[0022] In various embodiments, the content sources 175 include broadcast television and radio sources, video on demand platforms and streaming video and audio services platforms, one or more content data networks, data servers, web servers and other content servers, and/or other sources of media.

[0023] In various embodiments, the communications network 125 can include wired, optical and/or wireless links and the network elements 150, 152, 154, 156, etc. can include service switching points, signal transfer points, service control points, network gateways, media distribution hubs, servers, firewalls, routers, edge devices, switches and other network nodes for routing and controlling communications traffic over wired, optical and wireless links as part

of the Internet and other public networks as well as one or more private networks, for managing subscriber access, for billing and network management and for supporting other network functions.

[0024] FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a system functioning within the communication network of FIG. 1 in accordance with various aspects described herein. In one or more embodiments, system 200 can include a location management server (LMS) 202a communicatively coupled to edge server 202g and edge server 202h over communication network 202b. Further, the edge server 202g can be communicatively coupled to router 202c and 202d over communication network 202b as well as edge server 202h can be communicatively coupled to router 202e and router 202f over communication network 202b. In addition, router 202c, router 202d, router 202e, and router 202f can be communicatively coupled to the LMS 202a over communication network 202b. Also, communication device 202i associated with the user 202k, and communication device 202j and communication device 202q are associated with user 202l can be communicatively coupled to edge server 202g over a communication network, and to router 202c and router 202d over communication network 202b via edge server 202g. Further, communication device 202m associated with the user 202o and communication device 202n associated with user 202p can be communicatively coupled to edge server 202h over a communication network, and to router 202e and router 202f over communication network 202b via edge server 202h. Each of communication device 202i, communication device 202j, communication device 202m, communication device 202n, and communication device 202q can comprise a holographic communication device, desktop computer, television, set-top box, media processor, laptop computer, tablet computer, smartphone, mobile phone, mobile device, gaming console, virtual reality device, cross reality device, augmented reality device, or a combination thereof. Each of LMS 202a, edge server 202g, edge server 202h can comprise one or more servers in the same location, one or more servers in different locations, one or more virtual servers in the same location or in different locations, one or more cloud servers, or a combination thereof. Further, an edge server can provide support for applications running or executing on a communication device such as a holographic communication session, video game, virtual reality environment, etc. as well as manage network resources or network conditions applicable to the communication device such as bandwidth, latency, throughput, etc.

[0025] In one or more embodiments, the LMS 202s can determine that communication device 202i (e.g., holographic communication device) initiates a communication session with communication device 202m (e.g., holographic communication device) over communication network 202b. The LMS 202a can determine the initiation of the communication session by monitoring the data or information between communication device 202i and communication device 202m. In other embodiments, either communication device 202i or communication device 202m can send a notification to LMS 202a that indicates that a communication session has been initiated between communication device 202i and communication device 202m.

[0026] In one or more embodiments, the LMS 202a may want to verify the location of communication device 202m that is associated with user 202o to prevent possible mali-

cious users (e.g., hackers) impersonating a legitimate user. One way a malicious user can impersonate a legitimate user is to mask their location (e.g., with a VPN) and disguise it with a location associated with a legitimate user. This can be done in a case in which user 202k and communication device 202i is associated with a customer service department of a financial institution that can deal with sensitive information during a communication session. To verify the location of communication device, the LMS 202a can transmit a notification to communication device 202m that indicates to communication device 202m to initiate a location verification application that is stored on it (the location verification application can be within software associated with the communicating with the financial institution. Further, the location verification application on communication device 202m transmits a group of notifications to a group of adjacent network devices that can include edge server 202h, router 202e, and router 202f. Each of the notifications can include the IP address of the LMS 202a and an instruction to provide the respective location of each adjacent network device to the LMS 202a. In addition, the group of adjacent network devices including edge server 202h, router 202e, and router 202f can each transmit their respective locations to the LMS 202a.

[0027] In one or more embodiments, the LMS 202a can receive the group of locations from the group of adjacent network devices over communication network 202b including the edge server 202h, router 202e, and router 202f. Further, the LMS 202a can identify a location of the communication device 202m. In addition, the LMS 202a can determine that the location of the communication device 202m is in proximity (within a distance threshold) to group of locations associated with the adjacent network devices including the edge server 202h, router 202e, and router 202f, thereby verifying the location of communication device 202m. Also, in response to the location of the communication device 202m being verified, the LMS server 202a can establish a communication session between communication device 202i and communication device 202m over communication network 202b. In some embodiments, the location of communication device 202m can be encoded, encrypted, or otherwise stored, by the LMS 202a into a blockchain to be accessed by the LMS 202a at later time to verify the location of the communication device 202m during the current communication session or in a subsequent communication session. Encoding, encrypting, or otherwise storing the location of communication device 202m into a blockchain provides added security from malicious users (e.g., hackers) of fraudulently accessing and changing a stored location of the communication device 202m to perpetrate some fraud.

[0028] In one or more embodiments, (legitimate) user 202o of communication device 202m can be utilizing a VPN that masks the location of communication device 202m. Further, the communication device 202m can provide its location via user-generated input provided by user 202o. However, if the communication session is to be used to discuss confidential or sensitive matters, the user 202k associated with communication device 202i may like to verify the location provided by user 202o is in proximity to the location of adjacent network devices including edge server 202h, router 202e, and router 202f utilizing the embodiments described herein to ensure no malicious user

(e.g., hacker) is masking the location of communication device **202m** to perpetrate some fraud.

[0029] In one or more embodiments, the LMS **202a** can receive user-generated input indicating an alleged location of communication device **202m** as well as a location information from adjacent network devices of communication device **202m** that indicate that the user **202o** is not in proximity (within a distance threshold) of their residence, the location of which is stored in a database and accessible by the LMS **202a** or a blockchain. Further, the user **202k**, communication device **202i**, and LMS **202a** can be associated with the financial institution that has access to financial transaction information of user **202o**. The LMS server **202a** can access and analyze the financial transaction information (e.g., purchase transaction made by a credit card, debit card, etc.) to determine that the user **202o** has traveled to the location indicated in the user-generated input and in proximity to the location of the adjacent network devices including edge server **202h**, router **202e**, and router **202f**. The purchase transaction information can indicate airline tickets or hotel reservations that are in proximity to the purported location of the communication device **202m** as well as the group of adjacent network devices, thereby verifying the location of communication device **202m**.

[0030] In one or more embodiments, the LMS **202a** can request, periodically or at random time intervals, to receive the group of locations of the group of adjacent network devices to the communication device **202m** including edge server **202h**, router **202e**, and router **202f** to make sure any hacker did not mask the location of the communication device **202m** or the location of the adjacent network devices at the initiation of the communication session but failed to do so during the communication session. Further, the LMS **202a** receives the location of each of the group of adjacent network devices during the communication session resulting in a second group of locations and may determine that the location of the communication device **202m** is not in proximity to the second group of locations. Consequently, the LMS can terminate the communication session between communication device **202i** and communication device **202m**. In addition, the LMS **202a** can send a notification to communication device **202i** that the location of communication device **202m** has been detected to be fraudulent. Such a notification can be presented on the display of communication device **202i** for viewing by the user.

[0031] In one or more embodiments, the LMS **202a** can determine that communication device **202q** (e.g., holographic communication device) has initiated a communication session with communication device **202m**. The LMS **202a** can send a notification to communication device **202m** to instruct the location verification application on communication device **202m** to send a group of notifications to a group of adjacent network devices to communication device **202m** including edge server **202h**, router **202e**, and router **202f** to transmit their respective locations to the LMS **202a**. Further, the LMS **202a** can receive a group of locations of the adjacent network devices including the edge server **202h**, router **202e**, and router **202f**. In addition, the LMS **202a** can obtain the location of communication device **202m** from the blockchain. Also, the LMS **202a** can determine that the location of the communication device **202m** from the blockchain is in proximity (e.g., within a distance threshold) to each of the group of locations of the adjacent network devices. Consequently, the LMS can facilitate or allow a

communication session between communication device **202q** and communication device **202m**. In other embodiments, the LMS **202a** can determine that the location of the communication device **202** from the blockchain is not in proximity (e.g., not within a distance threshold) to any of the group of locations of the adjacent network devices indicating that the user **202o** is masking the location of communication device **202m** for malicious purposes. Thus, the LMS **202a** can cease the establishing of the communication session between communication device **202q** and communication device **202m**. Further, the LMS **202a** can send a notification to communication device **202q** that is presented to user **202l** to indicate that the location of communication device **202m** is fraudulent.

[0032] In one or more embodiments, the communication device **202q** can request the identity of user **202o** from the LMS **202a**. Further, the LMS **202a** can request the identity of user **202o** via communication device **202m** and can receive user-generated input from user **202o** via communication device **202m** indicating the identity of the user. However, based one or more criteria (e.g., age, user's need for anonymity, user type, device type, etc.) the LMS **202a** can determine whether to provide the identity of the user **202o** to communication device **202q**. If the LMS **202a** determines that the identity of user **202o** can be revealed to communication device **202q**, it provides a notification to communication device **202q** that does so and that is presented to user **202l**. If the LMS **202a** determines that the identity of user **202o** cannot be revealed to communication device **202q**, it provides a notification to communication device **202q** that is presented to user **202l** that indicates it cannot reveal the identity of user **202o**. For example, if user **202o** is a child and user **202l** is associated with a virtual field trip provider, revealing the identity of user **202o** would not be appropriate. In an additional example, the user type associated with user **202o** can indicate a user younger than 18 years of age such that revealing the identity of the user **202o** would not be appropriate. In a further example, the device type associated with communication device **202m** can indicate that it is a gaming console associated with children such that revealing the identity of user **202o** would not be appropriate.

[0033] In one or more embodiments, the LMS **202a** can determine that communication device **202j** initiates a communication session with communication device **202n** over communication network **202b**. The LMS **202a** can determine the initiation of the communication session by monitoring the data or information between communication device **202j** and communication device **202n**. In other embodiments, either communication device **202j** or communication device **202n** can send a notification to LMS **202a** that indicates that a communication session has been initiated between communication device **202j** and communication device **202n**.

[0034] In one or more embodiments, the LMS **202a** can transmit a notification to communication device **202n** that indicates to communication device **202n** to initiate a location application that is stored on it. Further, the location verification application on communication device **202n** transmits a group of notifications to a group of adjacent network devices that can include edge server **202h**, router **202e**, and router **202f**. In addition, the group of adjacent network

devices including edge server **202h**, router **202e**, and router **202f** can each transmit their respective locations to the LMS **202a**.

[0035] In one or more embodiments, the LMS **202a** can receive the group of locations from the group of adjacent network devices over communication network **202b** including the edge server **202h**, router **202e**, and router **202f**. Further, the LMS **202a** can identify a location of the communication device **202n**. In addition, the LMS **202a** can determine that the location of the communication device **202n** is in proximity (e.g., within a distance threshold) to group of locations associated with the adjacent network devices including the edge server **202h**, router **202e**, and router **202f**. Also, in response to verifying the location of communication device **202n** with its adjacent network devices, the LMS server **202a** can establish a communication session between communication device **202j** and communication device **202n** over communication network **202b**. In some embodiments, the location of communication device **202n** can be encoded, encrypted, or otherwise stored, by the LMS **202a**, into a blockchain to be accessed by the LMS **202a** at later time to verify the location of the communication device **202n** during the current communication session or in a subsequent communication session.

[0036] FIGS. 2B-2D depicts illustrative embodiments of methods in accordance with various aspects described herein. Referring to FIG. 2B, in one or more embodiments, aspects of method **210** can be performed by an LMS, a communication device, or one or more adjacent network devices including, but not limited to, an edge server and one or more routers. The method **210** can include the LMS, at **210a**, determining a first communication device initiating a first communication session with a second communication device over a communication network. Further, the method **210** can include the LMS, at **210c**, transmitting a first notification to the second communication device. The first notification indicates to the second communication device to initiate a location verification application on the second communication device. In addition, the method **210** can include the second communication device, by the location verification application, at **210d**, transmitting a group of notifications to a group of adjacent network devices. Each notification of the group of notifications indicates to an adjacent network device of the group of adjacent network devices to provide its location to the LMS. Also, the method **210** can include the LMS, at **210e**, receiving a first group of locations from the group of adjacent network devices over the communication network. Further, the method **210** can include the LMS, at **210f**, identifying a first location associated with the second communication device (e.g., the second communication device provides the LMS with its location via user-generated input). In addition, the method **210** can include the LMS, at **210g**, determining that the first location is in proximity (e.g., within a distance threshold) to the first group of locations. Also, the method **210** can include the LMS, at **210h**, establishing the first communication session between the first communication device and the second communication device over the communication network.

[0037] In one or more embodiments, the first communication session comprises a holographic communication session. Further, the method **210** can include the LMS, at **210b**, determining a virtual private network (VPN) is associated

with the second communication device that masks the location of the location of the second communication device.

[0038] In one or more embodiments, the method **210** can include the LMS, at **210i**, receiving a second group of locations from the group of adjacent network devices during the first communication session. Further, the method **210** can include the LMS, at **210j**, determining that the first location is not in proximity (e.g., not within a distance threshold) to the second group of locations. In addition, the method **210** can include the LMS, at **210k**, terminating the first communication session between the first communication device and the second communication device. Also, the method **210** can include the LMS, at **210l**, transmitting a second notification to the first communication device that indicates that a second location associated with the second communication device is fraudulent. Further, the method **210** can include the LMS, at **210m**, encoding, encrypting, or otherwise storing the first location into a blockchain.

[0039] Referring to FIG. 2C, in one or more embodiments, aspects of method **220** can be performed by an LMS, a communication device or one or more adjacent network devices including, but not limited to, an edge server and one or more routers. The method **220** can include the LMS, at **220a**, determining a first communication device initiating a first communication session with a second communication device over a communication network. Further, the method **220** can include the LMS, at **220c**, transmitting a first notification to the second communication device. The first notification indicates to the second communication device to initiate a location verification application on the second communication device. In addition, the method **220** can include the second communication device, by the location verification application, at **220d**, transmitting a group of notifications to a group of adjacent network devices. Each notification of the group of notifications indicates to an adjacent network device of the group of adjacent network devices to provide its location to the processing system. Also, the method **220** can include the LMS, at **220e**, receiving a first group of locations from the group of adjacent network devices over the communication network. Further, the method **220** can include the LMS, at **220f**, obtaining a first location associated with the second communication device from a blockchain. In addition, the method **220** can include the LMS, at **220g**, determining that the first location is in proximity (e.g., within a distance threshold) to the first group of locations. Also, the method **220** can include the LMS, at **220h**, establishing the first communication session between the first communication device and the second communication device over the communication network.

[0040] In one or more embodiments, the method **220** can include the LMS, at **220i**, determining that the first location is not in proximity (e.g., not within a distance threshold) to the first group of locations. Further, the method **220** can include the LMS, at **220j**, terminating the first communication session between the first communication device and the second communication device. In addition, the method **220** can include the LMS, at **220k**, transmitting a second notification to the first communication device that indicates that a second location associated with the second communication device is fraudulent.

[0041] Referring to FIG. 2D, in one or more embodiments, aspects of method **230** can be performed by an LMS. The

method **230** can include the LMS, at **230a**, receiving a request for an identity of a user associated with the second communication device from the first communication device. Further, the method **230** can include the LMS, at **230b**, identifying the user associated with the second communication device resulting in an identification of the user. In addition, the method **230** can include the LMS, at **230c**, determining to provide the identity of the user to the first communication device based on a criterion and the identification of the user resulting in a first determination. Also, the method **230** can include the LMS, at **230d**, providing a third notification to the first communication device that indicates the identity of the user based on the first determination. The third notification indicates the identity of the user. Further, the method **230** can include the LMS, at **230e**, determining not to provide the identity of the user to the first communication device based on a criterion and the identification of the user resulting in a second determination. In addition, the method **230** can include the LMS, at **230f**, providing a fourth notification to the first communication device that indicates the identity of the user cannot be revealed based on the second determination. The criterion comprises one of age, user type, device type, or a combination thereof.

[0042] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIGS. 2B-2D, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein. Further one or more blocks can be performed in response to one or more other blocks.

[0043] In addition, portions of some embodiments can be combined with portions of other embodiments.

[0044] Referring now to FIG. 3, a block diagram **300** is shown illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein. In particular a virtualized communication network is presented that can be used to implement some or all of the subsystems and functions of system **100**, the subsystems and functions of system **200**, and methods **210**, **220**, **230** presented in FIGS. 1, 2A, 2B, 2C, 2D, and 3. For example, virtualized communication network **300** can facilitate in whole or in part verifying location of a communication device with the location of adjacent network devices to initiate a communication session.

[0045] In particular, a cloud networking architecture is shown that leverages cloud technologies and supports rapid innovation and scalability via a transport layer **350**, a virtualized network function cloud **325** and/or one or more cloud computing environments **375**. In various embodiments, this cloud networking architecture is an open architecture that leverages application programming interfaces (APIs); reduces complexity from services and operations; supports more nimble business models; and rapidly and seamlessly scales to meet evolving customer requirements including traffic growth, diversity of traffic types, and diversity of performance and reliability expectations.

[0046] In contrast to traditional network elements—which are typically integrated to perform a single function, the virtualized communication network employs virtual net-

work elements (VNEs) **330**, **332**, **334**, etc. that perform some or all of the functions of network elements **150**, **152**, **154**, **156**, etc. For example, the network architecture can provide a substrate of networking capability, often called Network Function Virtualization Infrastructure (NFVI) or simply infrastructure that is capable of being directed with software and Software Defined Networking (SDN) protocols to perform a broad variety of network functions and services. This infrastructure can include several types of substrates. The most typical type of substrate being servers that support Network Function Virtualization (NFV), followed by packet forwarding capabilities based on generic computing resources, with specialized network technologies brought to bear when general purpose processors or general purpose integrated circuit devices offered by merchants (referred to herein as merchant silicon) are not appropriate. In this case, communication services can be implemented as cloud-centric workloads.

[0047] As an example, a traditional network element **150** (shown in FIG. 1), such as an edge router can be implemented via a VNE **330** composed of NFV software modules, merchant silicon, and associated controllers. The software can be written so that increasing workload consumes incremental resources from a common resource pool, and moreover so that it's elastic: so the resources are only consumed when needed. In a similar fashion, other network elements such as other routers, switches, edge caches, and middle-boxes are instantiated from the common resource pool. Such sharing of infrastructure across a broad set of uses makes planning and growing infrastructure easier to manage.

[0048] In an embodiment, the transport layer **350** includes fiber, cable, wired and/or wireless transport elements, network elements and interfaces to provide broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or access to content sources **175** for distribution of content to any or all of the access technologies. In particular, in some cases a network element needs to be positioned at a specific place, and this allows for less sharing of common infrastructure. Other times, the network elements have specific physical layer adapters that cannot be abstracted or virtualized, and might require special DSP code and analog front-ends (AFEs) that do not lend themselves to implementation as VNEs **330**, **332** or **334**. These network elements can be included in transport layer **350**.

[0049] The virtualized network function cloud **325** interfaces with the transport layer **350** to provide the VNEs **330**, **332**, **334**, etc. to provide specific NFVs. In particular, the virtualized network function cloud **325** leverages cloud operations, applications, and architectures to support networking workloads. The virtualized network elements **330**, **332** and **334** can employ network function software that provides either a one-for-one mapping of traditional network element function or alternately some combination of network functions designed for cloud computing. For example, VNEs **330**, **332** and **334** can include route reflectors, domain name system (DNS) servers, and dynamic host configuration protocol (DHCP) servers, system architecture evolution (SAE) and/or mobility management entity (MME) gateways, broadband network gateways, IP edge routers for IP-VPN, Ethernet and other services, load balancers, distributors and other network elements. Because these elements don't typically need to forward large amounts of traffic, their workload can be distributed across a number of servers—each of which adds a portion of the capability, and

overall which creates an elastic function with higher availability than its former monolithic version. These virtual network elements 330, 332, 334, etc. can be instantiated and managed using an orchestration approach similar to those used in cloud compute services.

[0050] The cloud computing environments 375 can interface with the virtualized network function cloud 325 via APIs that expose functional capabilities of the VNEs 330, 332, 334, etc. to provide the flexible and expanded capabilities to the virtualized network function cloud 325. In particular, network workloads may have applications distributed across the virtualized network function cloud 325 and cloud computing environment 375 and in the commercial cloud, or might simply orchestrate workloads supported entirely in NFV infrastructure from these third party locations.

[0051] Turning now to FIG. 4, there is illustrated a block diagram of a computing environment in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. 4 and the following discussion are intended to provide a brief, general description of a suitable computing environment 400 in which the various embodiments of the subject disclosure can be implemented. In particular, computing environment 400 can be used in the implementation of network elements 150, 152, 154, 156, access terminal 112, base station or access point 122, switching device 132, media terminal 142, and/or VNEs 330, 332, 334, etc. Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other program modules and/or as a combination of hardware and software. For example, computing environment 400 can facilitate in whole or in part verifying location of a communication device with the location of adjacent network devices to initiate a communication session. Further, each of LMS 202a, router 202c, router 202d, router, 202e, router 202f, edge server 202g, edge server 202h, communication device 202i, communication device 202j, communication device 202m, communication device 202n, and communication device 202q can comprise computing environment 400.

[0052] Generally, program modules comprise routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0053] As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

[0054] The illustrated embodiments of the embodiments herein can be also practiced in distributed computing envi-

ronments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0055] Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

[0056] Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0057] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0058] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0059] With reference again to FIG. 4, the example environment can comprise a computer 402, the computer 402 comprising a processing unit 404, a system memory 406 and a system bus 408. The system bus 408 couples system components including, but not limited to, the system memory 406 to the processing unit 404. The processing unit 404 can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as the processing unit 404.

[0060] The system bus 408 can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available

bus architectures. The system memory **406** comprises ROM **410** and RAM **412**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **402**, such as during startup. The RAM **412** can also comprise a high-speed RAM such as static RAM for caching data.

[0061] The computer **402** further comprises an internal hard disk drive (HDD) **414** (e.g., EIDE, SATA), which internal HDD **414** can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) **416**, (e.g., to read from or write to a removable diskette **418**) and an optical disk drive **420**, (e.g., reading a CD-ROM disk **422** or, to read from or write to other high capacity optical media such as the DVD). The HDD **414**, magnetic FDD **416** and optical disk drive **420** can be connected to the system bus **408** by a hard disk drive interface **424**, a magnetic disk drive interface **426** and an optical drive interface **428**, respectively. The hard disk drive interface **424** for external drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0062] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **402**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0063] A number of program modules can be stored in the drives and RAM **412**, comprising an operating system **430**, one or more application programs **432**, other program modules **434** and program data **436**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **412**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0064] A user can enter commands and information into the computer **402** through one or more wired/wireless input devices, e.g., a keyboard **438** and a pointing device, such as a mouse **440**. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often connected to the processing unit **404** through an input device interface **442** that can be coupled to the system bus **408**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

[0065] A monitor **444** or other type of display device can be also connected to the system bus **408** via an interface,

such as a video adapter **446**. It will also be appreciated that in alternative embodiments, a monitor **444** can also be any display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer **402** via any communication means, including via the Internet and cloud-based networks. In addition to the monitor **444**, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

[0066] The computer **402** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **448**. The remote computer(s) **448** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements described relative to the computer **402**, although, for purposes of brevity, only a remote memory/storage device **450** is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) **452** and/or larger networks, e.g., a wide area network (WAN) **454**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0067] When used in a LAN networking environment, the computer **402** can be connected to the LAN **452** through a wired and/or wireless communication network interface or adapter **456**. The adapter **456** can facilitate wired or wireless communication to the LAN **452**, which can also comprise a wireless AP disposed thereon for communicating with the adapter **456**.

[0068] When used in a WAN networking environment, the computer **402** can comprise a modem **458** or can be connected to a communications server on the WAN **454** or has other means for establishing communications over the WAN **454**, such as by way of the Internet. The modem **458**, which can be internal or external and a wired or wireless device, can be connected to the system bus **408** via the input device interface **442**. In a networked environment, program modules depicted relative to the computer **402** or portions thereof, can be stored in the remote memory/storage device **450**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

[0069] The computer **402** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0070] Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks

use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10BaseT wired Ethernet networks used in many offices.

[0071] Turning now to FIG. 5, an embodiment 500 of a mobile network platform 510 is shown that is an example of network elements 150, 152, 154, 156, and/or VNEs 330, 332, 334, etc. For example, platform 510 can facilitate in whole or in part verifying location of a communication device with the location of adjacent network devices to initiate a communication session. In one or more embodiments, the mobile network platform 510 can generate and receive signals transmitted and received by base stations or access points such as base station or access point 122. Generally, mobile network platform 510 can comprise components, e.g., nodes, gateways, interfaces, servers, or disparate platforms, that facilitate both packet-switched (PS) (e.g., internet protocol (IP), frame relay, asynchronous transfer mode (ATM)) and circuit-switched (CS) traffic (e.g., voice and data), as well as control generation for networked wireless telecommunication. As a non-limiting example, mobile network platform 510 can be included in telecommunications carrier networks, and can be considered carrier-side components as discussed elsewhere herein. Mobile network platform 510 comprises CS gateway node(s) 512 which can interface CS traffic received from legacy networks like telephony network(s) 540 (e.g., public switched telephone network (PSTN), or public land mobile network (PLMN)) or a signaling system #7 (SS7) network 560. CS gateway node(s) 512 can authorize and authenticate traffic (e.g., voice) arising from such networks. Additionally, CS gateway node(s) 512 can access mobility, or roaming, data generated through SS7 network 560; for instance, mobility data stored in a visited location register (VLR), which can reside in memory 530. Moreover, CS gateway node(s) 512 interfaces CS-based traffic and signaling and PS gateway node(s) 518. As an example, in a 3GPP UMTS network, CS gateway node(s) 512 can be realized at least in part in gateway GPRS support node(s) (GGSN). It should be appreciated that functionality and specific operation of CS gateway node(s) 512, PS gateway node(s) 518, and serving node(s) 516, is provided and dictated by radio technology (ies) utilized by mobile network platform 510 for telecommunication over a radio access network 520 with other devices, such as a radiotelephone 575.

[0072] In addition to receiving and processing CS-switched traffic and signaling, PS gateway node(s) 518 can authorize and authenticate PS-based data sessions with served mobile devices. Data sessions can comprise traffic, or content(s), exchanged with networks external to the mobile network platform 510, like wide area network(s) (WANs) 550, enterprise network(s) 570, and service network(s) 580, which can be embodied in local area network(s) (LANs), can also be interfaced with mobile network platform 510 through PS gateway node(s) 518. It is to be noted that WANs 550 and enterprise network(s) 570 can embody, at least in part, a service network(s) like IP multimedia subsystem (IMS). Based on radio technology layer(s) available in

technology resource(s) or radio access network 520, PS gateway node(s) 518 can generate packet data protocol contexts when a data session is established; other data structures that facilitate routing of packetized data also can be generated. To that end, in an aspect, PS gateway node(s) 518 can comprise a tunnel interface (e.g., tunnel termination gateway (TTG) in 3GPP UMTS network(s) (not shown)) which can facilitate packetized communication with disparate wireless network(s), such as Wi-Fi networks.

[0073] In embodiment 500, mobile network platform 510 also comprises serving node(s) 516 that, based upon available radio technology layer(s) within technology resource(s) in the radio access network 520, convey the various packetized flows of data streams received through PS gateway node(s) 518. It is to be noted that for technology resource(s) that rely primarily on CS communication, server node(s) can deliver traffic without reliance on PS gateway node(s) 518; for example, server node(s) can embody at least in part a mobile switching center. As an example, in a 3GPP UMTS network, serving node(s) 516 can be embodied in serving GPRS support node(s) (SGSN).

[0074] For radio technologies that exploit packetized communication, server(s) 514 in mobile network platform 510 can execute numerous applications that can generate multiple disparate packetized data streams or flows, and manage (e.g., schedule, queue, format . . .) such flows. Such application(s) can comprise add-on features to standard services (for example, provisioning, billing, customer support . . .) provided by mobile network platform 510. Data streams (e.g., content(s) that are part of a voice call or data session) can be conveyed to PS gateway node(s) 518 for authorization/authentication and initiation of a data session, and to serving node(s) 516 for communication thereafter. In addition to application server, server(s) 514 can comprise utility server(s), a utility server can comprise a provisioning server, an operations and maintenance server, a security server that can implement at least in part a certificate authority and firewalls as well as other security mechanisms, and the like. In an aspect, security server(s) secure communication served through mobile network platform 510 to ensure network's operation and data integrity in addition to authorization and authentication procedures that CS gateway node(s) 512 and PS gateway node(s) 518 can enact. Moreover, provisioning server(s) can provision services from external network(s) like networks operated by a disparate service provider; for instance, WAN 550 or Global Positioning System (GPS) network(s) (not shown). Provisioning server(s) can also provision coverage through networks associated to mobile network platform 510 (e.g., deployed and operated by the same service provider), such as the distributed antennas networks shown in FIG. 1(s) that enhance wireless service coverage by providing more network coverage.

[0075] It is to be noted that server(s) 514 can comprise one or more processors configured to confer at least in part the functionality of mobile network platform 510. To that end, the one or more processor can execute code instructions stored in memory 530, for example. It is should be appreciated that server(s) 514 can comprise a content manager, which operates in substantially the same manner as described hereinbefore.

[0076] In example embodiment 500, memory 530 can store information related to operation of mobile network platform 510. Other operational information can comprise

provisioning information of mobile devices served through mobile network platform **510**, subscriber databases; application intelligence, pricing schemes, e.g., promotional rates, flat-rate programs, couponing campaigns; technical specification(s) consistent with telecommunication protocols for operation of disparate radio, or wireless, technology layers; and so forth. Memory **530** can also store information from at least one of telephony network(s) **540**, WAN **550**, SS7 network **560**, or enterprise network(s) **570**. In an aspect, memory **530** can be, for example, accessed as part of a data store component or as a remotely connected memory store.

[0077] In order to provide a context for the various aspects of the disclosed subject matter, FIG. 5, and the following discussion, are intended to provide a brief, general description of a suitable environment in which the various aspects of the disclosed subject matter can be implemented. While the subject matter has been described above in the general context of computer-executable instructions of a computer program that runs on a computer and/or computers, those skilled in the art will recognize that the disclosed subject matter also can be implemented in combination with other program modules. Generally, program modules comprise routines, programs, components, data structures, etc. that perform particular tasks and/or implement particular abstract data types.

[0078] Turning now to FIG. 6, an illustrative embodiment of a communication device **600** is shown. The communication device **600** can serve as an illustrative embodiment of devices such as data terminals **114**, mobile devices **124**, vehicle **126**, display devices **144** or other client devices for communication via either communications network **125**. For example, communication device **600** can facilitate in whole or in part verifying location of a communication device with the location of adjacent network devices to initiate a communication session. Further, each of LMS **202a**, router **202c**, router **202d**, router, **202e**, router **202f**, edge server **202g**, edge server **202h**, communication device **202i**, communication device **202j**, communication device **202m**, communication device **202n**, and communication device **202q** can comprise communication device **600**.

[0079] The communication device **600** can comprise a wireline and/or wireless transceiver **602** (herein transceiver **602**), a user interface (UI) **604**, a power supply **614**, a location receiver **616**, a motion sensor **618**, an orientation sensor **620**, and a controller **606** for managing operations thereof. The transceiver **602** can support short-range or long-range wireless access technologies such as Bluetooth®, ZigBee®, WiFi, DECT, or cellular communication technologies, just to mention a few (Bluetooth® and ZigBee® are trademarks registered by the Bluetooth® Special Interest Group and the ZigBee® Alliance, respectively). Cellular technologies can include, for example, CDMA-1X, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, SDR, LTE, as well as other next generation wireless communication technologies as they arise. The transceiver **602** can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VoIP, etc.), and combinations thereof.

[0080] The UI **604** can include a depressible or touch-sensitive keypad **608** with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device **600**. The keypad **608** can be an integral part of a housing

assembly of the communication device **600** or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth®. The keypad **608** can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI **604** can further include a display **610** such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **600**. In an embodiment where the display **610** is touch-sensitive, a portion or all of the keypad **608** can be presented by way of the display **610** with navigation features.

[0081] The display **610** can use touch screen technology to also serve as a user interface for detecting user input. As a touch screen display, the communication device **600** can be adapted to present a user interface having graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The display **610** can be equipped with capacitive, resistive or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements or other functions of the user interface. The display **610** can be an integral part of the housing assembly of the communication device **600** or an independent device communicatively coupled thereto by a tethered wireline interface (such as a cable) or a wireless interface.

[0082] The UI **604** can also include an audio system **612** that utilizes audio technology for conveying low volume audio (such as audio heard in proximity of a human ear) and high volume audio (such as speakerphone for hands free operation). The audio system **612** can further include a microphone for receiving audible signals of an end user. The audio system **612** can also be used for voice recognition applications. The UI **604** can further include an image sensor **613** such as a charged coupled device (CCD) camera for capturing still or moving images.

[0083] The power supply **614** can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and/or charging system technologies for supplying energy to the components of the communication device **600** to facilitate long-range or short-range portable communications. Alternatively, or in combination, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port or other suitable tethering technologies.

[0084] The location receiver **616** can utilize location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device **600** based on signals generated by a constellation of GPS satellites, which can be used for facilitating location services such as navigation. The motion sensor **618** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect motion of the communication device **600** in three-dimensional space. The orientation sensor **620** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **600** (north, south, west, and east, as well as combined orientations in degrees, minutes, or other suitable orientation metrics).

[0085] The communication device **600** can use the transceiver **602** to also determine a proximity to a cellular, WiFi, Bluetooth®, or other wireless access points by sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or signal time of arrival (TOA) or time of flight (TOF) measurements. The controller **606** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), programmable gate arrays, application specific integrated circuits, and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies for executing computer instructions, controlling, and processing data supplied by the aforementioned components of the communication device **600**.

[0086] Other components not shown in FIG. 6 can be used in one or more embodiments of the subject disclosure. For instance, the communication device **600** can include a slot for adding or removing an identity module such as a Subscriber Identity Module (SIM) card or Universal Integrated Circuit Card (UICC). SIM or UICC cards can be used for identifying subscriber services, executing programs, storing subscriber data, and so on.

[0087] The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and doesn’t otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

[0088] In the subject specification, terms such as “store,” “storage,” “data store,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can comprise both volatile and nonvolatile memory, by way of illustration, and not limitation, volatile memory, non-volatile memory, disk storage, and memory storage. Further, non-volatile memory can be included in read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can comprise random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SL-DRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to comprise, without being limited to comprising, these and any other suitable types of memory.

[0089] Moreover, it will be noted that the disclosed subject matter can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments where tasks are performed by

remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0090] In one or more embodiments, information regarding use of services can be generated including services being accessed, media consumption history, user preferences, and so forth. This information can be obtained by various methods including user input, detecting types of communications (e.g., video content vs. audio content), analysis of content streams, sampling, and so forth. The generating, obtaining and/or monitoring of this information can be responsive to an authorization provided by the user. In one or more embodiments, an analysis of data can be subject to authorization from user(s) associated with the data, such as an opt-in, an opt-out, acknowledgement requirements, notifications, selective authorization based on types of data, and so forth.

[0091] Some of the embodiments described herein can also employ artificial intelligence (AI) to facilitate automating one or more features described herein. The embodiments (e.g., in connection with automatically identifying acquired cell sites that provide a maximum value/benefit after addition to an existing communication network) can employ various AI-based schemes for carrying out various embodiments thereof. Moreover, the classifier can be employed to determine a ranking or priority of each cell site of the acquired network. A classifier is a function that maps an input attribute vector, $x=(x_1, x_2, x_3, x_4, \dots, x_n)$, to a confidence that the input belongs to a class, that is, $f(x)=\text{confidence}(\text{class})$. Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to determine or infer an action that a user desires to be automatically performed. A support vector machine (SVM) is an example of a classifier that can be employed. The SVM operates by finding a hypersurface in the space of possible inputs, which the hypersurface attempts to split the triggering criteria from the non-triggering events. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches comprise, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein also is inclusive of statistical regression that is utilized to develop models of priority.

[0092] As will be readily appreciated, one or more of the embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing UE behavior, operator preferences, historical information, receiving extrinsic information). For example, SVMs can be configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to pre-determined criteria which of the acquired cell sites will benefit a maximum number of subscribers and/or which of the acquired cell sites will add minimum value to the existing communication network coverage, etc.

[0093] As used in some contexts in this application, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

[0094] Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

[0095] In addition, the words “example” and “exemplary” are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to

present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

[0096] Moreover, terms such as “user equipment,” “mobile station,” “mobile,” “subscriber station,” “access terminal,” “terminal,” “handset,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a subscriber or user of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings.

[0097] Furthermore, the terms “user,” “subscriber,” “customer,” “consumer” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based, at least, on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

[0098] As employed herein, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

[0099] As used herein, terms such as “data storage,” “data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components or computer-readable storage media, described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory.

[0100] What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of com-

ponents or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible. Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

[0101] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0102] As may also be used herein, the term(s) “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner than can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

[0103] Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that have not been described in the subject disclosure. Further,

more than or less than all of the features described with respect to an embodiment can also be utilized.

What is claimed is:

1. A device, comprising:

a processing system including a processor; and

a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising:

determining a first communication device initiating a livestreaming communication session with a second communication device over a communication network;

transmitting a group of notifications to a group of adjacent network devices, via the second communication device, wherein each notification of the group of notifications indicates to an adjacent network device of the group of adjacent network devices to provide its location to the processing system;

receiving a first group of locations from the group of adjacent network devices over the communication network;

identifying a first location associated with the second communication device;

determining that the first location is in proximity to the first group of locations resulting in a first determination; and

establishing the livestreaming communication session between the first communication device and the second communication device over the communication network based on the first determination.

2. The device of claim 1, wherein the determining of the first location is in proximity to the first group of locations comprises that determining that the first location is within a distance threshold of a second location of the first group of locations.

3. The device of claim 1, wherein the determining of the first location is in proximity to the first group of locations comprises that determining that the first location is within a distance threshold of each location of the first group of locations.

4. The device of claim 1, wherein the first location is encrypted, wherein the operations comprise decrypting the first location.

5. The device of claim 1, wherein the first location is encrypted in a blockchain, wherein the operations comprise decrypting the first location from the blockchain.

6. The device of claim 1, wherein the operations comprise determining whether the first location is in proximity to the first group of locations during the communication session resulting in a second determination.

7. The device of claim 6, wherein the operations comprise:

determining that the first location is not in proximity to the first group of locations during the communication session based on the second determination resulting in a third determination; and

discontinuing the communication session based on the third determination.

8. The device of claim 6, wherein the operations comprise:

determining that the first location is in proximity to the first group of locations during the communication session based on the second determination resulting in a fourth determination; and

continuing the communication session based on the fourth determination.

9. The device of claim 6, wherein the determining of the first location is in proximity to the first group of locations during the communication session comprises determining the first location is in proximity to the first group of locations during the communication session based on a periodic time interval.

10. The device of claim 6, wherein the determining of the first location is in proximity to the first group of locations during the communication session comprises determining the first location is in proximity to the first group of locations during the communication session based on a random time interval.

11. A non-transitory, machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising:

determining a first communication device initiating a livestreaming communication session with a second communication device over a communication network; transmitting a group of notifications to a group of adjacent network devices, via the second communication device, wherein each notification of the group of notifications indicates to an adjacent network device of the group of adjacent network devices to provide its location to the processing system;

receiving a first group of locations from the group of adjacent network devices over the communication network;

obtaining a first location associated with the second communication device;

decrypting the first location associated with the second communication device;

determining that the first location is in proximity to the first group of locations resulting in a first determination; and

establishing the livestreaming communication session between the first communication device and the second communication device over the communication network based on the first determination.

12. The non-transitory, machine-readable medium of claim 11, wherein the determining of the first location is in proximity to the first group of locations comprises that determining that the first location is within a distance threshold of a second location of the first group of locations.

13. The non-transitory, machine-readable medium of claim 11, wherein the determining of the first location is in proximity to the first group of locations comprises that determining that the first location is within a distance threshold of each location of the first group of locations.

14. The non-transitory, machine-readable medium of claim 11, wherein the first location is encrypted in a blockchain, wherein decrypting the first location comprises decrypting the first location from the blockchain.

15. The non-transitory, machine-readable medium of claim 11, wherein the operations comprise determining

whether the first location is in proximity to the first group of locations during the communication session resulting in a second determination.

16. The non-transitory, machine-readable medium of claim 15, the operations comprise:

determining that the first location is not in proximity to the first group of locations during the communication session based on the second determination resulting in a third determination; and

discontinuing the communication session based on the third determination.

17. The non-transitory, machine-readable medium of claim 15, determining that the first location is in proximity to the first group of locations during the communication session based on the second determination resulting in a fourth determination; and

continuing the communication session based on the fourth determination.

18. The non-transitory, machine-readable medium of claim 15, wherein the determining of the first location is in proximity to the first group of locations during the communication session comprises determining the first location is in proximity to the first group of locations during the communication session based on a periodic time interval.

19. The non-transitory, machine-readable medium of claim 15, wherein the determining of the first location is in proximity to the first group of locations during the communication session comprises determining the first location is in proximity to the first group of locations during the communication session based on a random time interval.

20. A method, comprising:

determining, by a processing system including a processor, a first communication device initiating a livestreaming communication session with a second communication device over a communication network;

transmitting, by the processing system, a group of notifications to a group of adjacent network devices, via the second communication device, wherein each notification of the group of notifications indicates to an adjacent network device of the group of adjacent network devices to provide its location to the processing system;

receiving, by the processing system, a first group of locations from the group of adjacent network devices over the communication network;

identifying, by the processing system, a first location associated with the second communication device;

determining, by the processing system, that the first location is within a distance threshold of a second location of the first group of locations resulting in a first determination; and

establishing, by the processing system, the livestreaming communication session between the first communication device and the second communication device over the communication network based on the first determination.

* * * * *